THE AFFINE STRATIFICATION NUMBER OF A NOETHERIAN SCHEME IS BOUNDED BY ITS DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 21613-21636, immediately following the statement at 21607-21611.

Editorial note. The reason for dim(Z) < dim(X) (source line 21629 asserts it without proof) is recorded in proof_roadmap as the reviewer's checklist; the author text is retained verbatim.

Editorial note. All pointers are cross-chapter (Properties x3, Morphisms x1), recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Dimension zero Noetherian schemes are affine. Properties, Lemma properties-lemma-locally-Noetherian-dimension-0, cited at proof lines 21612-21613. Base case d = 0 of the induction.
- Generic points of irreducible components. Properties, Lemma properties-lemma-Noetherian-irreducible-components, cited at proof lines 21614-21616. Lists the finitely many generic points eta_1,...,eta_n that the affine opens must contain.
- Affine opens through a point and its generalizations. Properties, Lemma properties-lemma-point-and-maximal-points-affine, cited at proof lines 21616-21618 and 21622-21624. Gives affine opens containing all the generic points, used twice (cover step and intersection step).
- Affineness detected on an affine cover. Morphisms, Lemma morphisms-lemma-characterize-affine, cited at proof lines 21625-21627. U -> X is affine because U -> U_j is affine for every member of the finite affine cover.

Source reference. Original reference 01S8: focused-seeds/geometry/S9-morph-asn-bound-noetherian/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BA8: focused-seeds/geometry/S9-morph-asn-bound-noetherian/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AAX: focused-seeds/geometry/S9-morph-asn-bound-noetherian/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZX: focused-seeds/geometry/S9-morph-asn-bound-noetherian/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a Noetherian scheme of dimension $\infty > d \geq 0$. Then the affine stratification number of X is at most d.*

Proof. By induction on d. If $d = 0$, then X is affine, see Properties, Lemma 0AAX. Assume $d > 0$. Let $\eta_1, \ldots, \eta_n$ be the generic points of the irreducible components of X (Properties, Lemma 0BA8). We can cover X by affine opens containing $\eta_1, \ldots, \eta_n$, see Properties, Lemma 01ZX. Since X is quasi-compact we can find a finite affine open covering $X = \bigcup_{j=1,\ldots,m} U_j$ with $\eta_1, \ldots, \eta_n \in U_j$ for all $j = 1, \ldots, m$. Choose an affine open $U \subset U_1 \cap \ldots \cap U_m$ containing $\eta_1, \ldots, \eta_n$ (possible by the lemma already quoted). Then the morphism $U \to X$ is affine because $U \to U_j$ is affine for all j, see Morphisms, Lemma 01S8. Let $Z = X \setminus U$. By construction $\dim(Z) < \dim(X)$. By induction hypothesis we can find an affine stratification $Z = \bigcup_{i \in \{0,\ldots,n\}} Z_i$ of Z with $n \leq \dim(Z)$. Setting $U = X_{n+1}$ and $X_i = Z_i$ for $i \leq n$ we conclude. □